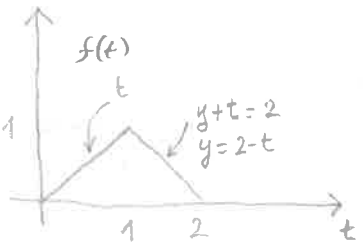
#1

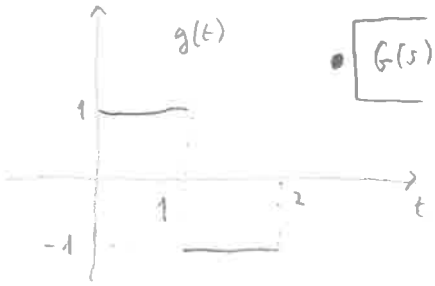


- $\boxed{F(s)} = \int_0^\infty f(t)\,e^{-st}\,dt = \int_0^1 t e^{-st}\,dt + \int_1^2 (2-t)e^{-st}\,dt$

$= \frac{1}{s^2} - \frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s} + \frac{1}{s^2}e^{-2s} + \frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s} = \boxed{\frac{1}{s^2}\left[1-2e^{-s}+e^{-2s}\right]}$

$\int te^{-st}dt = \left\{\begin{matrix} u=t & du=dt \\ dv=e^{-st} & v=-\frac{1}{s}e^{-st}\end{matrix}\right\} = -\frac{t}{s}e^{-st} + \frac{1}{s}\int e^{-st}dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}\Big]_0^1 = \frac{1}{s^2} - \frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s}$

$\int (2-t)e^{-st}dt = \left\{\begin{matrix} u=(2-t) & du=-dt \\ dv=e^{-st} & v=-\frac{1}{s}e^{-st}\end{matrix}\right\} = -(2-t)\frac{1}{s}e^{-st} - \frac{1}{s}\int e^{-st} = -(2-t)\frac{1}{s}e^{-st} + \frac{1}{s^2}e^{-st}\Big]_1^2 = \frac{1}{s^2}e^{-2s} + \frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s}$



- $\boxed{G(s)} = \int_0^1 e^{-st}dt - \int_1^2 e^{-st}dt = \left[-\frac{1}{s}e^{-st}\right]_0^1 + \left[\frac{1}{s}e^{-st}\right]_1^2$

$= \frac{1}{s} - \frac{1}{s}e^{-s} + \frac{1}{s}e^{-2s} - \frac{1}{s}e^{-s}$

$= \boxed{\frac{1}{s}\left[1-2e^{-s}+e^{-2s}\right]}$

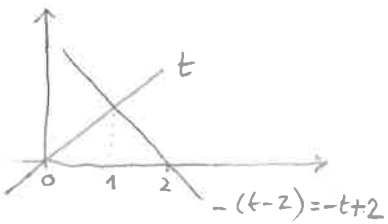
- Ja que $g(t) = f'(t) \Rightarrow \boxed{G(s)} = sF(s) - f(0)$

$= \frac{1}{s}\left[1-2e^{-s}+e^{-2s}\right] - 0 = \boxed{\frac{1}{s}\left[1-2e^{-s}+e^{-2s}\right]}$

Alternativament, $f(t) = t\cdot[u(t)-u(t-1)] + (-t+2)[u(t-1)-u(t-2)]$

$= tu(t) - (t+t-2)u(t-1) + (t-2)u(t-2)$

$= tu(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$



$\Rightarrow F(s) = \frac{1}{s^2} - 2e^{-s}\frac{1}{s^2} + \frac{1}{s^2}e^{-2s}$

#3

$\left.\begin{matrix} x'' = x + 2y \\ y' = \delta(t-3) * \sin(4t)\end{matrix}\right\}$ $\left.\begin{matrix} x(0)=0,\ x'(0)=1 \\ y(0)=0\end{matrix}\right\}$

Fent transformades de Laplace, ens queda

(si $x(t) \longrightarrow X(s)$, aleshores $x'(t) \longrightarrow sX(s)-x(0)$ & $x''(t) \longrightarrow s^2X(s)-sx(0)-x'(0)$)

$\left.\begin{matrix} s^2X(s) - 1 = X(s) + 2Y(s) \\ sY(s) = e^{-3s}\cdot\frac{4}{s^2+16}\end{matrix}\right\}$ D'on $X(s)(1-s^2) = -1-2Y(s)$; $X(s) = \frac{1+2Y(s)}{s^2-1}$

$\boxed{Y(s) = e^{-3s}\frac{4}{s(s^2+16)}}$

$\boxed{X(s) = \frac{1+\frac{e^{-3s}\cdot 8}{s(s^2+16)}}{s^2-1} = \frac{s(s^2+16)+8e^{-3s}}{s(s^2+16)(s^2-1)}}$

(#4) Solució general de $y'' - 8y' + 20y = e^t \cos t$

polinomi caract. $r^2 - 8r + 20 = 0$; $r = \frac{8 \pm \sqrt{64-80}}{2} = 4 \pm \frac{1}{2}\sqrt{-16} = 4 \pm 2j$

soluc. de la homogènia $y_H(t) = A e^{4t} \cos 2t + B e^{4t} \sin 2t$; $\alpha, \beta \in \mathbb{R}$

soluc. particular $y_p(t) = \alpha e^t \cos t + \beta e^t \sin t$

$y_p'(t) = \alpha e^t \cos t - \alpha e^t \sin t + \beta e^t \sin t + \beta e^t \cos t$

$y_p''(t) = \alpha e^t \cos t - \alpha e^t \sin t - \alpha e^t \sin t - \alpha e^t \cos t + \beta e^t \sin t + \beta e^t \cos t + \beta e^t \cos t - \beta e^t \sin t$

$$e^t \sin t(-2\alpha + 8\alpha - 8\beta + 20\beta) + e^t \cos t(2\beta - 8\alpha - 8\beta + 20\alpha) = e^t \cos t$$

$$\left.\begin{array}{l} 6\alpha + 12\beta = 0 \\ 12\alpha - 6\beta = 1 \end{array}\right\} \begin{array}{l} \alpha + 2\beta = 0 \; ; \; \alpha = -2\beta \\ -24\beta - 6\beta = 1 \; ; \; \left(\beta = -\frac{1}{30}\right) \left(\alpha = \frac{2}{30} = \frac{1}{15}\right) \end{array}$$

Per tant, $y_p(t) = \frac{e^t}{30}(2\cos t - \sin t)$

i la solució general és $\boxed{y(t) = e^{4t}(A \cos 2t + B \sin 2t) + \frac{e^t}{30}(2\cos t - \sin t)}$

(#2) Resoleu, mitjançant transformada de Laplace,

$$y'' - 2y' + 4y = e^t + \delta(t) \; ; \; y(0) = y'(0) = 0$$

Tenim que $s^2 Y(s) - 2sY(s) + 4Y(s) = \frac{1}{s-1} + 1 = \frac{s}{s-1}$; $Y(s)[s^2 - 2s + 4] = \frac{s}{s-1}$

Per tant, $Y(s) = \frac{s}{(s^2 - 2s + 4)(s-1)}$

$= \frac{Ms + N}{(s-1)^2 + 3} + \frac{A}{s-1}$

$s^2 - 2s + 4 = 0$;

$s = \frac{2 \pm \sqrt{4-16}}{2} = \frac{2 \pm 2\sqrt{3}j}{2}$

$= 1 \pm j\sqrt{3}$

$(s - 1 - j\sqrt{3})(s - 1 + j\sqrt{3}) = (s-1)^2 + 3$

$\Rightarrow Ms^2 - Ms + Ns - N + As^2 - 2As + A + 3A = s$

$$\Rightarrow \left.\begin{array}{l} M + A = 0 \\ -M + N - 2A = 1 \\ -N + 4A = 0 \end{array}\right\} \begin{array}{l} A = -M \\ N + M = 1 \\ -N - 4M = 0 \end{array} \Big\} -3M = 1 \; ; \; \left(M = -\frac{1}{3}\right) \left(A = \frac{1}{3}\right) \left(N = 1 - M = 1 + \frac{1}{3} = \frac{4}{3}\right)$$

Així, $Y(s) = \frac{1}{3}\left[\frac{-s+4}{(s-1)^2+3} + \frac{1}{s-1}\right] = \frac{1}{3}\left[\frac{-(s-1)}{(s-1)^2+3} + \frac{3}{(s-1)^2+3} + \frac{1}{s-1}\right]$ ($3 = (\sqrt{3})^2$)

$\xrightarrow{\mathcal{L}^{-1}}$ $\boxed{y(t) = \frac{1}{3}e^{-t}\left[\cos\sqrt{3}t + \sqrt{3}\sin\sqrt{3}t + 1\right]}$